

PARSA KAMALIPOUR

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RESEARCH INTERESTS

- Graph Theory & Graph Algorithms
- Design & Analysis of Algorithms
- Community Detection
- Social Network Analysis
- Complex Networks
- Machine Learning on Graphs

EDUCATION

Concordia University

Master of Computer Science (Thesis-based), advised by Prof. Hovhannes Harutyunyan

Thesis: *Geodesic-Compact Community Detection and Quality Assessment in Social Networks*

Graduate Coursework: Advanced Analysis of Algorithms, Combinatorial Algorithms, Algorithm Design Techniques, Machine Learning

Sep 2024 – 2027 (expected)

Montréal, QC, Canada

Vali-e-Asr University of Rafsanjan

B.Sc. in Computer Engineering, advised by Dr. Fahimeh Dabaghi-Zarandi

Graduated with Honors · B.Sc. Thesis Grade: **20/20**

Thesis: *Community detection in complex networks via an improved randomized algorithm* → published in the *Journal of Network and Computer Applications* (Q1)

Sep 2018 – Jun 2023

Rafsanjan, Iran

PUBLICATIONS

Journal Articles

- [1] *Spider Community Detection: Seeded Geodesic Expansion with Modularity-Guided Refinement and Greedy Merge Matching*. H. Harutyunyan, **P. Kamalipour** (2026). *Computers*, 15(2), 83. [DOI]
- [2] *Community Detection in Complex Networks Based on an Improved Random Algorithm Using Local and Global Network Information*. F. Dabaghi-Zarandi, **P. Kamalipour** (2022). *Journal of Network and Computer Applications* (Q1), vol. 206. [DOI] → Cited 25+ times.

Peer-Reviewed Conference Papers

- [3] *From Dense Graphs to Meaningful Communities: Assessing Community Quality Using Geodesic Distance Modularity on Metric Backbone-Sparsified Networks*. **P. Kamalipour**, H. Harutyunyan (2025). *IEEE SNAMS 2025*, Vienna, Austria, pp. 234–241. [DOI]
- [4] *Extending Refactoring Detection to Kotlin: A Dataset and Comparative Study*. I. Hemati Moghadam, M. M. Afkhami, **P. Kamalipour**, V. Zaytsev (2024). *IEEE SANER 2024*, Rovaniemi, Finland, pp. 267–271. [DOI]

Under Review

- [5] *XlateRefactor: A Language-Agnostic Pipeline for Cross-Language Refactoring Detection Using LLM-Based Code Translation*. I. Hemati Moghadam, M. M. Afkhami, V. Zaytsev, M. H. Ashoori, H. Bazmandegan, F. Zarepour, **P. Kamalipour**. Under review at JSME (major revision).

RESEARCH EXPERIENCE

Graduate Research Assistant — Concordia University

Algorithms & Complexity Lab, Advisor: Prof. Hovhannes Harutyunyan

Aug 2024 – Present

Montréal, QC, Canada

- Designed the **Spider** community detection algorithm, combining seeded geodesic expansion, modularity-guided refinement, and greedy merge matching.
- Evaluated Spider against **Louvain**, **Leiden**, **Infomap**, **CNM**, and **Label Propagation** on nine real-world networks (34–8,035 vertices) and LFR benchmarks: top F1 and ARI on Karate Club and Political Blogs, and **NMI 0.88** on Primary School where flow and propagation-based baselines collapse to near-trivial partitions.
- Introduced **Weighted Average Geodesic Distance Modularity (wGDM)**, a size-consistent, label-free partition quality measure, and conducted the first systematic study of **Metric Backbone** sparsification with Leiden: **14–71%** edge reduction across seven networks while preserving, and on dense contact networks improving, detection quality (High School F1: 0.834 → 0.970).
- Built a fully reproducible experimental pipeline with fixed random seeds, baseline implementations, and automated evaluation scripts.

Research Collaborator (Remote) — University of Twente

Formal Methods and Tools (FMT) Group, Supervisor: Dr. Iman Hemati Moghadam

Aug 2023 – Mar 2024

Enschede, The Netherlands

- Implemented the **KotlinCode2Text** parser and integrated it into the **RefDetect** refactoring-detection framework; built two refactoring datasets used in the *SANER 2024* empirical evaluation.
- Developed and evaluated **LLM prompting strategies** for cross-language code translation, contributing to the **XlateRefactor** pipeline (241 commits, 71 repositories, ~2,000 validated refactorings across Java, Kotlin, and C), with all artifacts released publicly.

Undergraduate Research Assistant — Vali-e-Asr University of Rafsanjan

Department of Computer Engineering, Supervisor: Dr. Fahimeh Dabaghi-Zarandi

Aug 2021 – Mar 2024

Rafsanjan, Iran

- Developed the **CRLG** community detection framework based on a randomized algorithm exploiting local and global network information, with weighted probabilistic seeding and heuristic community merging.
- Evaluated on real networks and GN/LFR benchmarks, achieving up to **+10%** over LCDR, MOACO, Node2vec-SC, NE-N2V, CDASS, and TS in NMI, modularity, and density; published in *JNCA* (Q1).

TEACHING EXPERIENCE

Graduate Teaching Assistant — Concordia University

Gina Cody School of Engineering and Computer Science

Sep 2024 – Present

Montréal, QC, Canada

Recognition: **Dean's Excellence Award for Excellence in Teaching by a Teaching Assistant**, May 2026

- Accumulated **1,400+ hours** of tutorial instruction, laboratory demonstration, grading, and Programmer-On-Duty support across **11 courses** spanning theory, systems, and AI, including:
 - COMP 465: Design & Analysis of Algorithms
 - COMP 339: Combinatorics
 - COMP 335: Intro to Theoretical CS
 - COMP 472: Artificial Intelligence
 - COMP 233: Probability & Statistics for CS
 - SOEN 363: Data Systems for Software Eng.

Undergraduate Teaching Assistant — Vali-e-Asr University of Rafsanjan

Department of Computer Engineering

Mar 2021 – Jan 2024

Rafsanjan, Iran

- Served as **Head TA and tutorial leader** for foundational CS courses (Data Structures, Algorithms Design, Discrete Mathematics, AI, Data Mining), mentoring students and coordinating grading teams.

HONORS & AWARDS

- Dean's Excellence Award for Excellence in Teaching by a Teaching Assistant**, Concordia University May 2026
- Conference and Exposition Allowance**, Concordia University 2025
- DRW Graduate Scholarship in Computer Science**, Concordia University & DRW Company 2024
- Concordia Merit Scholarship** (Entrance Scholarship Award), School of Graduate Studies Sep 2024
- Financial Research Support (FRS)**, Gina Cody School of Engineering and Computer Science 2024
- Distinguished Student Award** — among all students of Vali-e-Asr University 2023
- Top Researcher Award** — among all undergraduate students in Kerman Province 2023

SELECTED PROJECTS

Scalable GNN-Based Community Detection (GitHub)

2024

- Built GCN and GraphSAGE pipelines with neighbor sampling and graph partitioning; reached **90%** accuracy on 10K-node SBM graphs, enabled training on Reddit where full-batch methods run out of memory, and analyzed accuracy–time–memory trade-offs for scalable GNN deployment.

Minimum-Cost Flow Algorithms on Randomized Networks (GitHub)

2024

- Implemented Successive Shortest Path from scratch and compared SSP, Capacity Scaling, Scaling-SSP, and Primal–Dual across 28 randomized Euclidean graph configurations; showed Primal–Dual consistently attains optimal cost while SSP degrades with density.

CONFERENCE PRESENTATIONS

IEEE SNAMS 2025 — Vienna, Austria

Nov 2025

- Oral presentation (in person): *“From Dense Graphs to Meaningful Communities: Assessing Community Quality Using Geodesic Distance Modularity on Metric Backbone-Sparsified Networks.”* [Slides]

ACADEMIC SERVICE & LEADERSHIP

- Sub-reviewer** for conference submissions, with Prof. Hovhannes Harutyunyan 2025 – Present
- Director, Research Assistant Committee** — Scientific Association of CE, Vali-e-Asr University 2022 – 2023
- Director, Teaching Assistant Committee** — Scientific Association of CE, Vali-e-Asr University 2022 – 2023
- Teaching Staff Member** — Vali-e-Asr Collegiate Programming Contest (VCPC) 2021 – 2022

TECHNICAL SKILLS

- Graph Mining & Network Science:** Community Detection, Link Prediction, Node Classification, Network Embeddings, Graph Algorithms, SNAP & LFR Benchmarks, Large-Scale Network Evaluation
- Programming:** Python, C, C++, C#, Java, MATLAB, Bash, Assembly (x86, ARM)
- ML & Data:** PyTorch, PyTorch Geometric, Scikit-learn, NetworkX, iGraph, NumPy, Pandas, SciPy, Matplotlib
- Tools & Databases:** Linux, Git, Docker, $\LaTeX$, Jupyter, PostgreSQL, MySQL, MongoDB, Neo4j

LANGUAGES

English: Proficient (C1) · **French:** Pre-intermediate (A2, in progress toward B2) · **Persian:** Native

REFERENCES

- Hovhannes Harutyunyan**, Professor, Concordia University hovhannes.harutyunyan@concordia.ca
Dept. of Computer Science & Software Engineering · Master's Advisor
- Iman Hemati Moghadam**, Researcher, Eindhoven University of Technology i.hemati.moghadam@tue.nl
Research Supervisor (SANER 2024 and XlateRefactor studies)
- Fahimeh Dabaghi-Zarandi**, Assistant Professor, Vali-e-Asr University of Rafsanjan f.dabaghi@vru.ac.ir
Dept. of Computer Engineering · Bachelor's Thesis Advisor